

Exploring Challenges and Opportunities in Developing Systems to Improve Shopping Addiction through Social Sensing Technology

Ling-Chung Liu
Yung-Han Kao
icevanilla@gapp.nthu.edu.tw
leokao0801@gapp.nthu.edu.tw
National Tsing Hua University
Hsinchu, Taiwan

Chien Wen (Tina) Yuan
tinayuan@ntnu.edu.tw
National Taiwan Normal University
Taipei, Taiwan

Nanyi Bi
nanyibi@ntu.edu.tw
National Taiwan University
Taipei, Taiwan

An-Nie Chung
Ming-Chyi Huang
DAX57@tpech.gov.tw
mch@tpech.gov.tw
Taipei City Hospital Songde Branch
Taipei, Taiwan

Chuang-Wen You
cwyou@mx.nthu.edu.tw
National Tsing Hua University
Hsinchu, Taiwan

ABSTRACT

Shopping addiction is a destructive behavioral addiction characterized by compulsive purchasing with negative consequences for one's finances and social well-being. In the current study, we conducted a literature review aimed at identifying difficulties in the self-monitoring of shopping behavior using existing technological solutions. Based on these results, we developed a guide for designing an online survey to explore the challenges and opportunities in applying social sensing to the problem of shopping addiction. The potential results of the online survey are expected to shed light on designing social sensing solutions to reduce shopping addiction.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Shopping Addiction; Social Sensing; Online Survey Study.

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1 INTRODUCTION

Shopping addiction (i.e., compulsive shopping disorder or onio-mania) can have a broad range of negative effects on individuals in terms of finances, interpersonal relationships, and well-being, thereby rendering the individual susceptible to depression [23]. A number of researchers in the field of addiction [18] have sought to discover the symptoms and root causes of shopping addiction as well as methods to resolve the issue. What's worse, it appears that this problem is no longer limited to offline activities. Online shopping platforms are rapidly becoming a convenient and affordable alternative to brick-and-mortar stores, thereby exacerbating the problem of shopping addiction.

However, there are few studies discussing how to properly incorporate the social sensing to assist in reducing shopping addiction. Christenson et al. [6] reported that 92% of compulsive shoppers are unable to curb their shopping habits; however, there has been relatively little research on the effectiveness of treatments developed specifically for shopping addiction [10]. Researchers in the fields of psychology [3] and neurology [9] have sought to identify the causes, symptoms, and behavioral manifestations of shopping addiction. Cognitive Behavioral Therapy (CBT) and various drugs [3] have been used to assist patients in coping with their cravings. A number of researchers have also sought to curb problematic behaviors through the use of technology [7, 22]; however, this approach has often exacerbated the problem by providing access to online markets (i.e., an additional avenue by which to indulge cravings). Note also that most existing technology-based solutions rely on the self-management capacity of the individual, yet many of whom experience mental health conditions co-occurring with shopping addiction [14].

Extensive research has demonstrated that one's commitment to recovery can be reinforced by providing access to people who are willing and able to provide guidance [8]. However, few studies have addressed the issue of incorporating social sensing to assist individuals in dealing with their shopping addiction. By social sensing,

various stakeholders who live closely or communicate frequently with the individuals with shopping addictions can contribute to collaborative efforts in understanding their thoughts and actions. Social sensing not only helps infer the risk of relapse to shopping addiction for introducing timely intervention but also offers an opportunity for these individual shoppers to leverage the data collected from related stakeholders in their social network to raise their awareness and give supporting resources for improving their shopping problems. Hence, we posited the following research questions based on a review of literature:

- What and how are the contextual factors associated with people with shopping addiction regarding their shopping addiction behavior?
- How do people with shopping addiction communicate with members in their social network regarding shopping addiction, and what barriers are likely to hinder their mutual understandings?
- What and how to bring in complementary support from the community to facilitate constructive communication and adjust compulsive shopping behaviors of people with shopping addiction?

In the current study, we plan to conduct an online survey aimed at identifying the challenges and opportunities associated with social collaborative sensing to deal with shopping addiction. This study has been divided into two phases. In the first phase, we will then select a group of candidates exhibiting a high tendency toward shopping addiction (i.e., BSAS score ≥ 12 [1]) with a good balance of demographic characteristics to enter the shopping addiction survey study in the second phase. These individuals will be given a structured shopping behavior questionnaire aimed at elucidating their attitudes and opinions, the use of technology, and communication with family members or other related stakeholders in their social network regarding shopping addiction. We are also putting together a follow-up questionnaire by which to probe global attitudes toward support structures (e.g., medical, family, friends, and other social groups). We expect that the responses will provide important guidance in the design of social sensing solutions, which contribute to shopping addiction recovery assistance.

2 RELATED WORK

2.1 Background of Shopping Addiction

The aim of shopping has superseded the acquisition of essential items, such as food, clothing, and entertainment. One's selection of personal items now plays a key role in the building of one's personal image and measuring one's quality of life. Many people also use shopping as a means to gain pleasure and/or mitigate negative emotions [21]. In other words, the decisions involved in selecting and acquiring objects give one a sense of control over who they are and what they can do [21]. Positive feedback mechanisms inherent in the process of acquiring and showing off new items result in many individuals becoming overly dependent on shopping, or called shopping addiction, otherwise known as compulsive buying disorder. Unlike ordinary consumers, individuals with shopping addiction pay less attention to the value and function of a product, and more to the process of shopping [17]. They also tend to engage in shopping as a way of letting themselves go. Note that at

present, the American Psychiatric Association (APA) has yet to include shopping addiction in the DSM-5 [2] of mental disorders [20]. Nonetheless, Maraz et al. [15] have clearly demonstrated the negative impacts of compulsive buying behavior. It has been estimated that shopping addiction in developed countries affects nearly 5% of adults. Symptoms of shopping addiction include a strong constant and urge to make purchases, feelings of frustration when unable to shop, and a loss of control over self-action. And, shopping addiction is often accompanied by other psychiatric disorders, such as eating disorders and impulse control disorders [23].

In the current study, *shopping addiction* is operationalized as the symptoms of a maladaptive preoccupation with buying or shopping with a negative impact on one's quality of life. Individuals with shopping addiction exhibit the following symptoms: (1) Frequent preoccupation with buying; (2) Purchase impulses described as irresistible, intrusive and/or senseless; (3) Frequent buying of more than what can be afforded; (4) The purchase of items that are not needed; (5) Spending more time than necessary engage in the process of shopping; (6) Purchasing decisions that leave the individual in economic difficulty (e.g., indebtedness or bankruptcy).

2.2 Self-monitoring Technology Assistance

E-commerce platforms on the internet have opened another avenue by which the individuals with shopping addiction can indulge their impulses [11]. Individuals who are prone to excessive online shopping are encouraged to manually unsubscribe from all promotions, uninstall or log out of shopping apps, and/or delete credit card information from their browser in order to make the purchase process somewhat inconvenient. Most existing technological solutions focus on systematizing the shopping process [12] or blocking access to shopping sites [13].

Other self-monitoring solutions are meant to encourage a healthier lifestyle (e.g., Habinator and Life-coach apps [7]) or enable mindfulness (e.g., Mindful Shopping [22]). The former apps make it easier for people with shopping addiction to create new life goals with rewards when those goals are achieved. Mindful Shopping [22] makes it easier for people with shopping addiction to create a practical shopping list, set a reasonable budget, and limit the duration of shopping sessions. It also asks the user questions aimed at promoting mindfulness. Manual solutions such as these are no doubt effective; however, they require that people with shopping addiction devote considerable effort to the self-reporting of data, which many individuals consider obtrusive. Researchers have also developed so-called friction technologies [16] aimed at helping users to deliberate over their choices. Moser et al. [16] identified the features that e-commerce sites use to encourage impulse buying and the tools that consumers can use to control online spending. Their work involved systematic content analysis of 200 top e-commerce websites in the U.S. as well as a survey of online impulse buying. The poor self-control ability of most people with shopping addiction renders them unable to follow most interventions for any length of time or seek interpersonal support (e.g., family or friends) [19].

2.3 Incorporating Interpersonal Help or Intervention

Addiction can have a profoundly negative effect on one's entire family. Nonetheless, family support can be a valuable component of the recovery process [24]. To leverage family support in the recovery process of addiction, researchers have explored alternative ways to design social sensing systems to collaboratively with patients to improve their problems of alcohol addiction [25] or smartphone addiction [5]. People with shopping addiction who have problems of self controlling over shopping behaviors, which is similar with those people with other types of addiction. So, they need someone with whom they can discuss their concerns, such as the members of a social network (e.g., family or close friends) to bring in interpersonal support to collaboratively improve people's shopping addiction. In cases where people with shopping addiction hide purchases from their family, social sensing solutions are needed to facilitate the sharing of personal purchase-related data. Individuals with a history of addiction are a particularly valuable resource in providing meaningful guidance. Such individuals can be found in self-help support groups, such as Shopaholics Anonymous (SA) [8]. Note, however, that such systems are generally conducted in person. To the best of our knowledge, no previous study has implemented technology-based interpersonal support systems for shopping addiction.

3 ONLINE SURVEY STUDY DESIGN

We are currently designing a two-stage online survey study to recruit people with shopping addictions and explore the challenges and opportunities in the use of social sensing to ameliorate shopping addiction.

3.1 Procedure

The fact that shopping addiction is not currently listed in the DSM-5 [2] of mental disorders due to insufficient evidence [20]. When individuals who face these difficulties of self-managing their shopping behaviors, they tend to not take them as seriously as addiction to substances like alcohol and drugs. To avoid over-pathologizing individuals' shopping behaviors, psychiatrists would follow the diagnosis criteria of DSM-5 to identify other mental disorders, e.g., anxiety or depression, which are usually comorbid to shopping addiction. Note that this also makes it difficult to recruit individuals with shopping addiction for a survey. We have therefore decided to implement the survey study in two phases including (1) a pre-study screening and (2) a shopping addiction survey study. Our plan is to first deploy the study on an online survey website and then recruit several hundred participants via email, social platforms, and in collaboration with psychiatrists from Taipei City Hospital.

3.1.1 Phase I – Pre-study Screening. At this point, our aim is to obtain participant consent, which will involve pressing an agree button after reviewing a short paragraph describing the voluntary nature of the study, the risks, benefits, and methods to maintain confidentiality. Because we want to recruit individuals experiencing difficulties in managing their shopping behavior, we need a screening tool to assess the severity of their shopping addiction. Therefore, we will then have the participants fill out a screening questionnaire

that includes demographic information and the Bergen Shopping Addiction Scale questionnaire (BSAS) [1] to assess the severity of their shopping addiction from seven addiction criteria (salience, mood modification, conflict, tolerance, withdrawal, relapse, and problems related to their problem shopping behaviors). Note that the BSAS questionnaire is divided into 7 subsections, each of which has a total of 4 questions based on a 5-level Likert scale. BSAS scores are obtained by taking the highest ranked item from each of seven 4-item clusters, each of which is given equal weight. In accordance with Andreassen et al. [1], we will include individuals with a cut-off score of ≥ 12 , which involves a score of 3 ("agree") or 4 ("completely agree") for at least four of the seven items.

From respondents with a tendency toward shopping addiction, we will collect demographic information (name, email, gender, age, and education) and key shopping profile (annual income) of the respondents. We estimate that it will take 5 to 10 minutes to complete the pre-study screening phase.

3.1.2 Phase II – Shopping Addiction Survey Study. From the collected responses, we will select a group of suitable candidates with the aim of ensuring a balance of demographics and key shopping profile. The second phase will be implemented in five parts, as follows: (1) Study introduction, (2) Shopping behavior survey, (3) Follow-up questionnaire, and (4) Demographic questionnaire. Most of the parts (i.e., study introduction, Bergen Shopping Addiction Scale questionnaire, and demographic questionnaire) are similar with the previous phase study but with more descriptions and questions specific for the phase two study. In the following, the design of the rest parts (i.e., shopping behavior survey, and follow-up questionnaire) are explained.

The Shopping Behavior Study asks participants to rate their attitudes and opinions towards their shopping behaviors or habits and the communication with others in their social network regarding shopping addiction. The design of survey questions was guided by the related questionnaires [1], and the literature and the research questions we reported in the previous section. The aim of the shopping behavior survey will be to obtain information related to the following sections of questions: (1) personal shopping preferences, (2) methods by which to identify shopping cues or to cope with problematic shopping, (3) attitudes toward overcoming shopping addiction including medical support, (4) the use of technology to cope with problematic shopping, and (5) communication with family members or other related stakeholders in their social network. The question sequence will be randomly and evenly presented with regard to the sections of questions. Participants will be prompted to answer a series of 7-point Likert scale questions ranging from "Strongly Agree" to "Strongly Disagree."

Participants will also be asked to complete a follow-up questionnaire probing global attitudes toward support structures, including medical interventions, family, friends, and other social groups. Participants will be asked to recount a story of how others helped them. They will be asked to identify the two support structures that are the most helpful and the two structures that are the least helpful. They will also be asked a number of open-ended questions on their concerns towards the design of social sensing technology solutions and what they found to be the most important features.

Completing the second phase of the survey is expected to take 30 to 40 minutes.

3.2 Planned Data Analysis

Survey data will be analyzed using an independent sample *t*-test (Chi-squared test) to compare groups with different demographic backgrounds (e.g., gender, age, income level) or different severity of their shopping addiction. Measures of participants' opinions or attitudes related to their shopping behaviors will be derived from the self-reported scores responded to the 7-point Likert scale questions. Statistical characteristics (e.g., mean or variance) of these values will be analyzed to identify any potential significant difference among those groups. Open-ended responses will undergo iterative coding independently by three experienced researchers with the aim of identifying salient themes [4].

4 EXPECTED CONTRIBUTIONS & CONCLUSION

This paper reviews literature to identify three critical research questions to understand potential challenges and opportunities in developing technology systems to improve shopping addiction through social sensing. Given the difficulties to recruit individuals with shopping addiction, we outline a research plan to conduct a two-stage online survey study. These individuals will be given a structured shopping behavior survey aimed at elucidating their attitudes and opinions related to their shopping behaviors, the use of technology to cope with problematic shopping, and communication with family members or other related stakeholders in their social network regarding shopping addiction.

We are currently working closely with psychiatrists to refine the design of the survey questionnaire. Upon receiving IRB approval, we will use online survey methods to recruit more individuals to fill out this online survey. However, we believe that the design of the online survey will help us collect results shedding the light on designing social sensing solutions to reduce shopping addiction and be able to stimulate discussions in the poster session.

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